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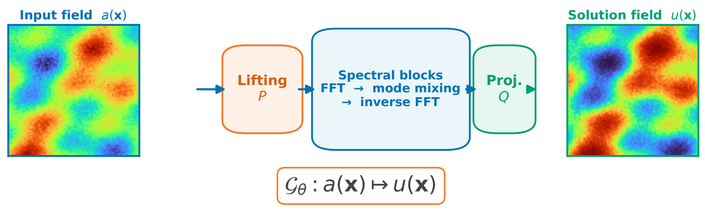
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Fourier Neural Operators for Engineering PDEs

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Learn the PDE solution operator, then evaluate it rapidly



FNO operator map

Input field → lifting → Fourier blocks → output field.

Introduction and objective

- Engineering PDEs describe heat transfer, fluid flow, porous-media transport, and coupled energy systems.
- Repeated high-fidelity solution can dominate uncertainty quantification, inverse analysis, optimization, and digital-twin updating.

Aim: Introduce FNOs through a reproducible synthetic 1D benchmark with numerical verification and physics-based diagnostics.

Operator formulation

- A well-posed parametric PDE and its boundary conditions define a solution operator.
- Complete input–output fields from a verified solver or experiment train $\mathcal{G}_\theta$ to predict new cases within the represented regime.

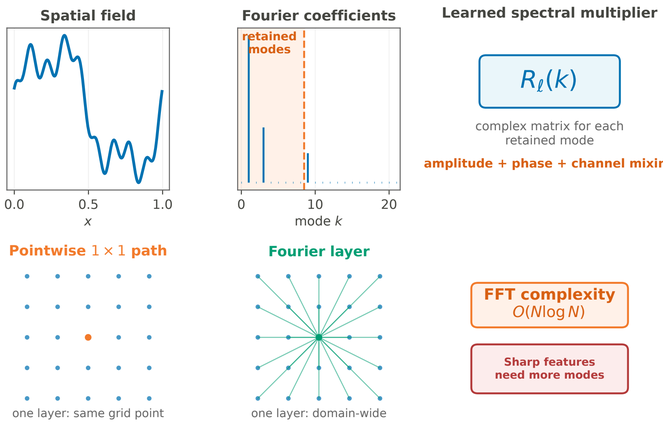
$$\mathcal{G}^\dagger : a(\mathbf{x}) \mapsto u(\mathbf{x}), \qquad \mathcal{G}_\theta \approx \mathcal{G}^\dagger$$

The object of learning is the complete solution map, not one scalar output or one fixed solution field.

$$\mathcal{D} = \left\{ (a^{(i)}, u^{(i)}) \right\}_{i=1}^{N_s}$$

Generic Fourier-space operator layer

$$v_{\ell+1}(\mathbf{x}) = \sigma\left[W_\ell v_\ell(\mathbf{x}) + \mathcal{F}^{-1}(R_\ell(\mathbf{k}) \mathcal{F}[v_\ell])(\mathbf{x}) + b_\ell \right]$$



Fourier mixing is global; retained modes set spectral resolution.

Spectral path

FFT → retain modes → apply complex matrices $R_\ell(\mathbf{k})$ → inverse FFT.

Pointwise path

Mixes latent channels at each grid point. The result is added to the spectral output before activation.

Prototype operator

$$\mathcal{G}_\theta : [\log k(x), x] \mapsto T(x)$$

The demonstrator solves steady heterogeneous heat conduction with fixed source and zero-temperature boundaries. Smooth random conductivity fields define the training distribution; layered fields are reserved for structured out-of-distribution testing.

Training data define the regime of reliability.

Primary takeaway

The reported errors are reproducible for this fixed synthetic benchmark, but flux, balance, positivity, and OOD checks show that field error alone is insufficient.

many simulations → one operator → rapid queries

Operator-learning workflow

A scientifically defensible workflow begins with a clearly defined operator and a verified reference solver. Fast inference is useful only after field accuracy, physical quantities, and generalization have been tested.

01 DEFINE THE OPERATOR — Inputs $\log k(x)$ and x; output $T(x)$. Fixed source and zero-Dirichlet boundaries define the admissible conductivity regime.

02 GENERATE VERIFIED FIELDS — 64-harmonic flux-form solver: 1000 train, 150 validation, 250 ID, and 250 layered-OOD fields. Verify analytic or quadrature references first.

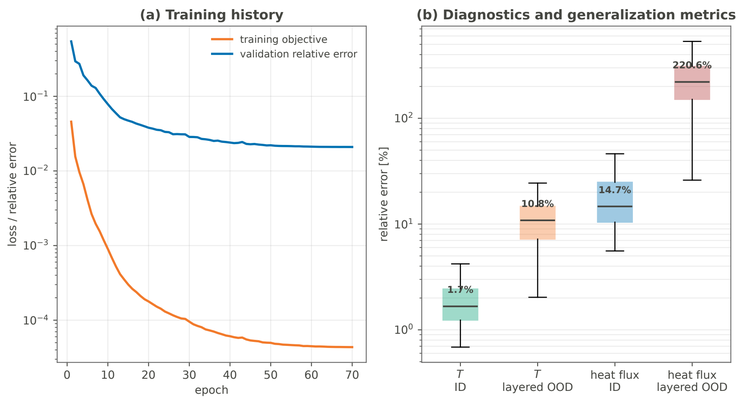
03 FIT THE FNO — Lift 2 to 32 channels; four residual Fourier blocks, 16 retained modes, and a pointwise path. Fit fields plus physics terms with AdamW for 70 epochs; keep the best validation checkpoint.

04 AUDIT THE SURROGATE — Audit temperature and face-flux errors, balance residual, boundary enforcement, and positivity. Compare smooth-ID with held-out layered-OOD distributions.

05 USE WITHIN THE TESTED REGIME — Repeated-query UQ, optimization, and inverse analysis belong inside the tested regime. Keep the reference solver for spot checks, OOD review, and recalibration.

Prototype: heterogeneous heat conduction

Prototype: 1D heterogeneous heat conduction



Temperature is easier than flux; layered fields are the hard OOD test.

Key findings

RESULT 1 For 250 unseen smooth conductivity fields, the median temperature error is **1.67%** (95th percentile: **3.92%**).

RESULT 2 The ID median heat-flux error is **14.68%**; boundary values are imposed exactly, but the median discrete-balance error is **12.52%**.

RESULT 3 Layered OOD medians rise to **10.85%** for temperature and **220.61%** for heat flux.

What the prototype evaluates

State field	Physics	Generalization
temperature profile $T_\theta(x)$	heat flux, boundary values, balance	smooth ID versus layered OOD

$$E_T = \frac{\|T_\theta - T\|_2}{\|T\|_2}, \qquad E_q = \frac{\|q_{h,\theta} - q_h\|_2}{\|q_h\|_2}$$

Limitations and next step

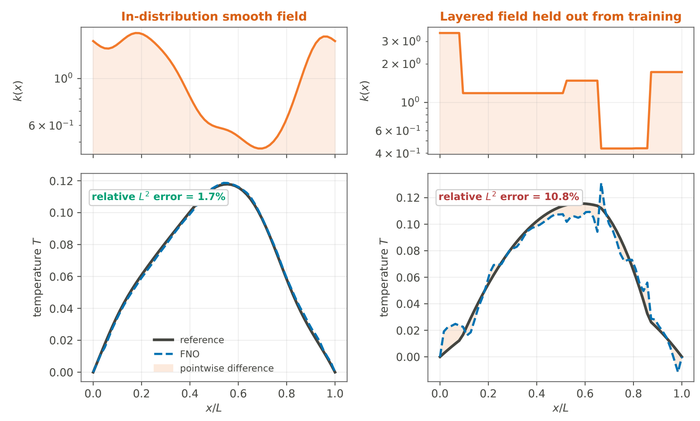
Single-seed synthetic 1D proof of concept with fixed source and fixed Dirichlet boundaries. It includes no experimental or field validation, measured speedup, baseline comparison, conservation enforcement, irregular geometry, multiphase flow, or long-time dynamics.

Next use: conservative global–local neural operators for porous-media and energy transport.

Primary result at a glance

On this fixed synthetic test, low in-distribution temperature error coexists with much larger derivative, balance, and structured out-of-distribution errors.

One trained operator: lower ID temperature error, much larger layered-OOD error



Representative median-error cases: smooth ID (left) and layered OOD (right).

1.67%	14.68%	10.85%	220.61%
ID median T	ID median flux	OOD median T	OOD median flux

Conclusions and emerging program

- The four principal error medians are reproducible from the archived fields and corrected checkpoint.
- Fourier layers provide domain-wide communication through efficient FFT-based spectral mixing.
- Low ID temperature error does not imply accurate flux, balance, positivity, or OOD behavior.
- Our emerging program targets physics-aware and conservative neural operators for energy and subsurface systems.

Practical implication

Use this as a reproducible numerical proof of concept, not field-scale validation. Require conservation, OOD, and reference-solver checks.

Why the direction matters

Subsurface / CCUS — permeability → pressure/saturation; UQ and well control

Thermal systems — rapid field prediction for design and optimization

Digital twins — fast updates, inverse analysis, and control

Implementation and reproducibility

1. Generate harmonic-face flux-form reference solutions for 1000 training and 150 validation fields.
2. Train a 1D FNO: 4 residual Fourier blocks, width 32, 16 modes; 72,033 tensor elements (137,569 real scalar DOF).
3. Evaluate temperature, heat flux, balance, positivity, boundaries, and smooth-ID versus layered-OOD behavior.

Fixed seed, corrected checkpoint, verification script, saved metrics, and complete generation code included.

Research direction and contact

Emerging research program. Physics-aware neural operators for porous-media flow, subsurface storage, heat and mass transport, pipeline systems, and multiscale energy models.

Prototype. 1000 training, 150 validation, 250 in-distribution test, and 250 layered OOD cases; fixed random seed 20260805.

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Contact / profile

Selected references. Li et al., *ICLR* (2021); Kovachki et al., *JMLR* **24**, 1–97 (2023); Li et al., Geo-FNO, *JMLR* **24**, 1–26 (2023).

Synthetic, single-seed computational results only. No experimental or field validation and no measured speedup are claimed. The package includes corrected weights, audit scripts, results, figures, and editable LaTeX.